

EXPERIMENT – 3

Performing Man-in-the-Middle (MITM) Attacks Using Ettercap in Kali Linux

Aim

To perform and analyze a Man-in-the-Middle (MITM) attack using the Ettercap tool in Kali Linux for educational and ethical hacking purposes.

Software Requirements

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Security Tool: Ettercap

System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support

Introduction

A Man-in-the-Middle (MITM) attack is a type of cyberattack where an attacker secretly intercepts and possibly alters communication between two parties without their knowledge.

Ettercap is a powerful open-source network security tool used for MITM attacks, traffic interception, and protocol analysis.

Using Kali Linux within Oracle VirtualBox provides a controlled and isolated environment to study MITM attacks ethically without affecting real-world systems.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Verify Network Connectivity

1. Ensure Kali Linux is connected to the same network as the target system.
2. Check IP address using:
3. `ifconfig`

Step 3: Launch Ettercap

1. Open Terminal in Kali Linux.
2. Start Ettercap with graphical interface:
3. `ettercap -G`

Step 4: Select Network Interface

1. In Ettercap GUI, click **Sniff Unified Sniffing**.
2. Select the active network interface (e.g., eth0 or wlan0).
3. Click **OK**.

Step 5: Scan for Hosts

1. Click **Hosts Scan for Hosts**.
2. After scanning, select **Hosts Hosts List**.
3. Identify the **target system** and **gateway**.

Step 6: Configure MITM Attack

1. Add the target system as **Target 1**.
2. Add the gateway/router as **Target 2**.
3. Click **MITM ARP Poisoning**.
4. Enable **Sniff remote connections**.
5. Click **OK**.

Step 7: Start Sniffing

1. Click **Start Start Sniffing**.
2. Observe intercepted traffic in real time.

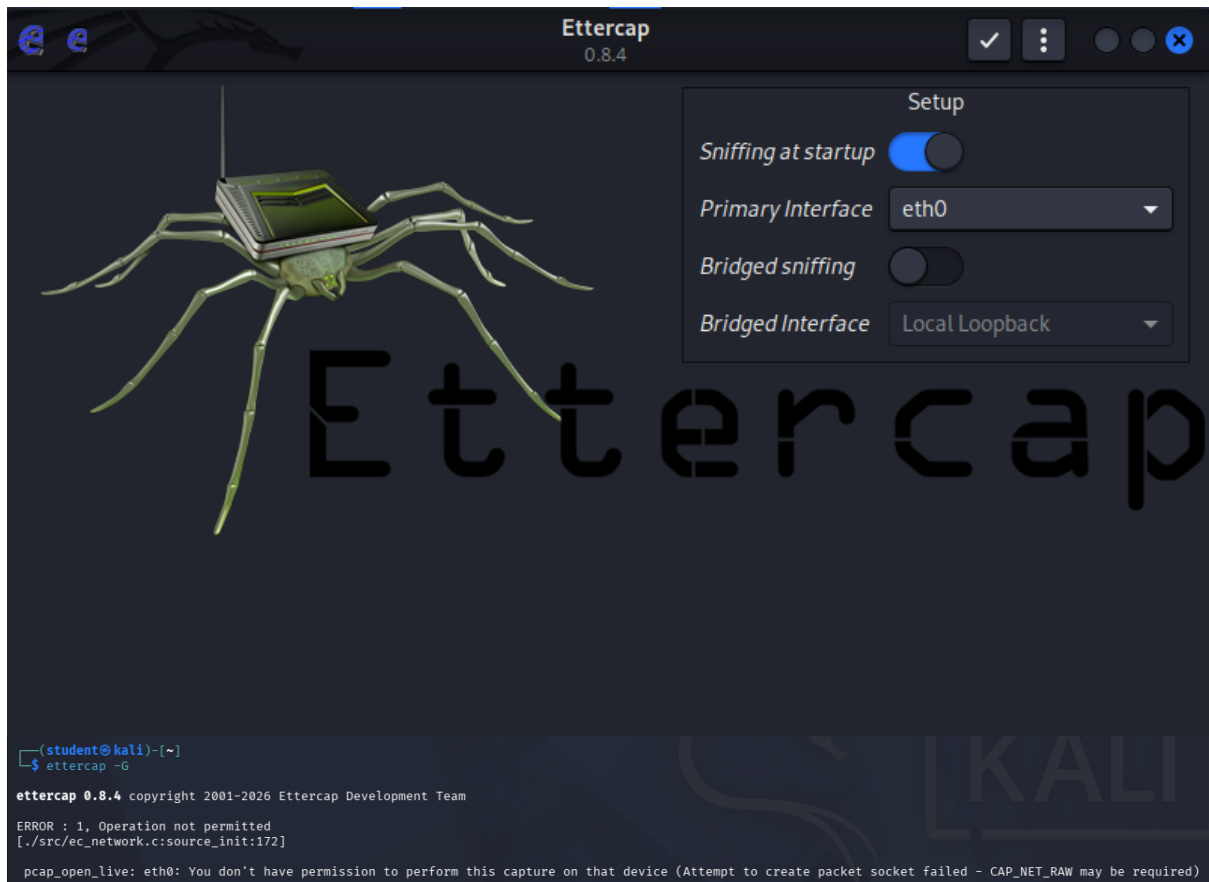
Output

```
(student@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:a00:27ff:fe30:fb7f prefixlen 64 scopeid 0<global>
    inet6 fd17:625c:f037:2:36ce:5aa1:b8d4:dafc prefixlen 64 scopeid 0<global>
    inet6 fe80::a00:27ff:fe30:fb7f prefixlen 64 scopeid 0<link>
    ether 08:00:27:30:fb:7f txqueuelen 1000 (Ethernet)
    RX packets 62478 bytes 75713694 (72.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 21199 bytes 10239962 (9.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(student@kali)-[~]
$ ettercap -G

ettercap 0.8.4 copyright 2001-2026 Ettercap Development Team
```



Network traffic between the target system and gateway is intercepted.

Credentials, packets, and communication details are visible through Ettercap.

Result

The MITM attack was successfully performed using Ettercap in Kali Linux. The experiment demonstrated how insecure network communication can be intercepted, highlighting the importance of encryption and secure network protocols.

Reference

<https://www.youtube.com/watch?v=GD2iWTn9HwU>

Note (Very Important for Records & Viva)

This experiment is performed **strictly for academic and ethical learning purposes** within a controlled lab environment.